CHAPTER 8

Portfolio Selection—Markowitz Model

Introduction

In his 1952 seminal article, Harry Markowitz stated that the objective of portfolio selection is to determine the allocation of securities in a portfolio such that it yields the maximum expected return given a specified risk or, alternatively, the minimum portfolio risk given a specified portfolio expected return. Given Equation (7.2) for the portfolio's expected return and Equation (7.11) for the portfolio variance, the Markowitz portfolio selection objective therefore involves determining the weights (w_i) for those equations that yield either the maximum $E(R_p)$ given a specified $V(R_p)$ or the minimum $V(R_p)$ given a specified $E(R_p)$. In this chapter, we examine the Markowitz portfolio selection model for determining optimum portfolio allocations. We begin our analysis by first examining the importance of correlation in determining the different return-risk combinations attainable by varying the security allocations of a two-stock portfolio. With this background, we next examine Markowitz portfolio selection and define an efficiency frontier in terms of that process. We conclude the chapter by extending the selection process to the single-index and multi-index models that were introduced in Chapter 6.

Two-Security Portfolio Return-Risk Relation

As we change the allocation of securities, we change both the portfolio's expected return and risk. How portfolio return and risk change in relation to each other depends on the correlations between the securities. This can be seen by examining the return-risk relationship of a two-security portfolio. Consider first a portfolio formed from

two perfectly positively correlated stocks A and B with the following expected returns and variances:

Stock A	Stock B
$E(r_A) = 12\%$	$E(r_B) = 18\%$
$V(r_A) = 16\%$	$V(r_B) = 36\%$
$\sigma(r_A) = 4\%$	$\sigma\ (r_B) = 6\%$
Correlation:	$Cov(r_A\ r_B) = 24$
	$\rho_{AB} = +1$

The return-risk relation of portfolios formed with these two stocks can be seen by varying the allocations of investment funds between the two stocks. For example, if all investment funds are placed in A ($w_A = 1$, $w_B = 0$), the portfolio return and standard deviation would be equal to A's expected return and standard deviation of 12 percent and 4 percent, respectively. If half of the funds are invested in A and half in B, then the portfolio expected return would be 15 percent and the portfolio standard deviation would be 5 percent:

$$E(R_p) = (0.5)(12\%) + (0.5)(18\%) = 15\%$$
$$\sigma(R_p) = \sqrt{(0.5)^2(16) + (0.5)^2(36) + 2(0.5)(0.5)(24)} = 5\%$$

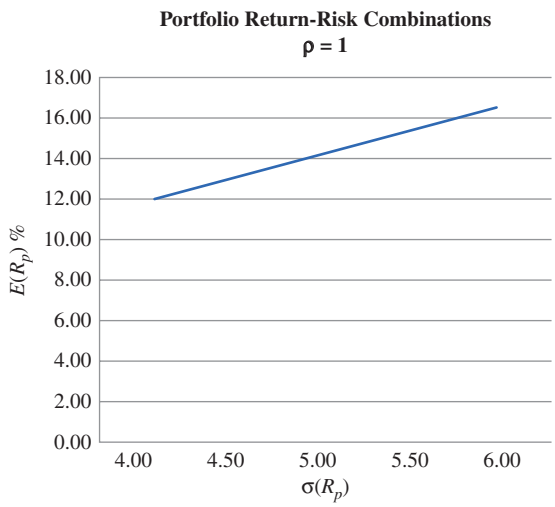
Finally, if all funds are place in B, then the portfolio return and risk would equal stock B's expected return and standard deviation of 18 percent and 6 percent, respectively. Exhibit 8.1 shows these and other portfolio return-risk combinations obtained by varying funds between A and B, and the figure in Exhibit 8.1 shows a graph of the relationship (note: the relation does not include any short positions; that is, all the weights are nonnegative).

The figure in Exhibit 8.1 shows a positive linear relationship between portfolio return and risk. The linear relationship suggests that the portfolio's return and risk are simply linear combinations of the return and risk of the two securities and do not depend on the correlation between securities. That is, when the correlation coefficient is one, the portfolio variance (or standard deviation) depends only on the securities' variances (or standard deviations). That is:

$$V(R_p) = w_1^2 V(r_1) + w_2^2 V(r_2) + 2w_1 w_2 Cov(r_1 r_2)$$
$$V(R_p) = w_1^2 V(r_1) + w_2^2 V(r_2) + 2w_1 w_2 \rho_{12}\sigma(r_1)\sigma(r_2)$$
$$V(R_p) = w_1^2 \sigma(r_1)^2 + w_2^2 \sigma(r_2)^2 + 2w_1 w_2 (1)\sigma(r_1)\sigma(r_2)$$
$$V(R_p) = [w_1\sigma(r_1) + w_2\sigma(r_2)]^2$$
$$\sigma(R_p) = [w_1\sigma(r_1) + w_2\sigma(r_2)]$$

EXHIBIT 8.1 Portfolio Return and Risk with Perfect Positive Correlation

Portfolio	w_A	w_B	$E(R_p)$	$\sigma\ (R_p)$
1	1	0	$E(R_A) = 12\%$	$\sigma\ (R_A) = 4$
2	0.75	0.25	13.5%	4.5
3	0.5	0.5	15.0%	5.0
4	0.25	0.75	16.5%	5.5
5	0	1	$E(R_B) = 18$	$\sigma\ (R_B) = 6$

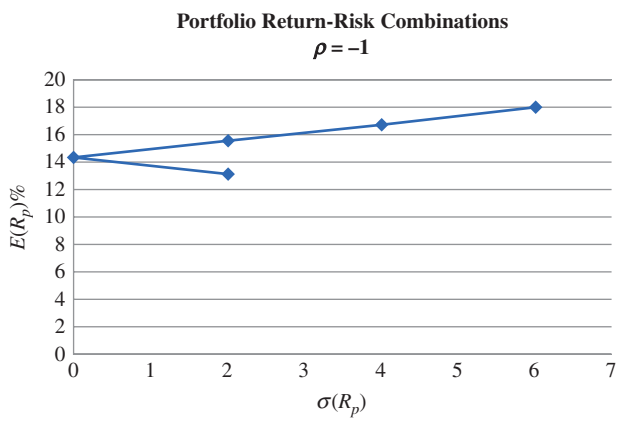


Intuitively, if two securities move in perfect unison with each other, there is no correlation benefit, and therefore the different portfolio return and risk combinations are linear combinations of the two securities' returns and risks.

Consider next the portfolio return-risk relation shown in Exhibit 8.2 where the returns of stocks A and B are assumed to be perfectly negatively correlated (−24 covariance). In this case, the portfolio return-risk relationship is characterized by two linear segments: a negatively sloped segment and a positively sloped segment. The negatively sloped segment extends from the 12 percent and 4 percent return-risk combination obtained by placing all funds in the low return-risk stock A to the 14.4 percent return and zero risk combination on the vertical axis obtained by investing 60 percent in A and 40 percent in B. The positively sloped segment extends from the vertical intercept to the 18 percent and 6 percent return-risk combination obtained by investing all funds in the high return-risk stock B. The positive-sloped portion of the figure includes all efficient portfolios and the negatively sloped portion consists of inefficient portfolios. *Efficient portfolios* are defined as those that yield the maximum return for a given risk, whereas inefficient portfolios are those that yield the minimum return for a given risk. Thus, at the 12 percent, 4 percent coordinate the return of 12 percent

EXHIBIT 8.2 Portfolio Return and Risk with Perfect Negative Correlation

Portfolio	w_A	w_B	$E(R_p)$	$\sigma\ (R_p)$
1	1	0	$E(R_A) = 12\%$	$\sigma\ (R_A) = 4$
2	0.8	0.2	13.2%	2
3	0.6	0.4	14.4%	0
4	0.4	0.6	15.6%	2
5	0.2	0.8	16.8%	4
6	0	1	$E(R_B) = 18\%$	$\sigma\ (R_B) = 6$



is the lowest return an investor can obtain for assuming a risk of 4. By changing the allocation from $w_A = 1$ and $w_B = 0$, to $w_A = 0.20$ and $w_B = 0.80$, the investor can move up to the positively sloped segment where for a risk of 4 percent the maximum return of 16.8 percent is obtained. Finally, note that the vertical intercept represents a zero risk portfolio. Whenever securities are perfectly negatively correlated, a graph of the portfolio's return-risk relation will always touch the vertical axis.[1]

In Exhibit 8.3, the return-risk relationships for both correlation cases are plotted on the same graph. The two curves define the limits within which all portfolios of these two securities must lie for any intermediate correlation coefficient between $\rho_{AB} = -1$ and $\rho_{AB} = +1$. To fit into the triangle ACB, the curve depicting the return-risk relation for an intermediate correlation has to be convex. This convex relation is illustrated in Exhibit 8.4, where the portfolio return-risk combinations are shown for the case in which the correlation coefficient for stocks A and B is zero. The convex shape of the portfolio return-risk curve means that for equal increases in the portfolio return, the portfolio risk as measured by the standard deviation will increase at an increasing rate. For example, as the portfolio return increases by equal increments of 1.2 percent, the portfolio standard deviation goes from 3.39 percent to 3.93 (change of 0.54), from 3.93 to 4.86 (change of 0.93), and from 4.86 to 6 (change of 1.14).

EXHIBIT 8.3 Portfolio Return and Risk with Different Correlations

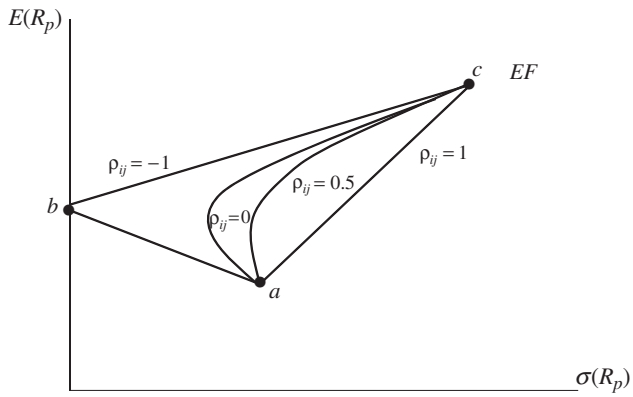
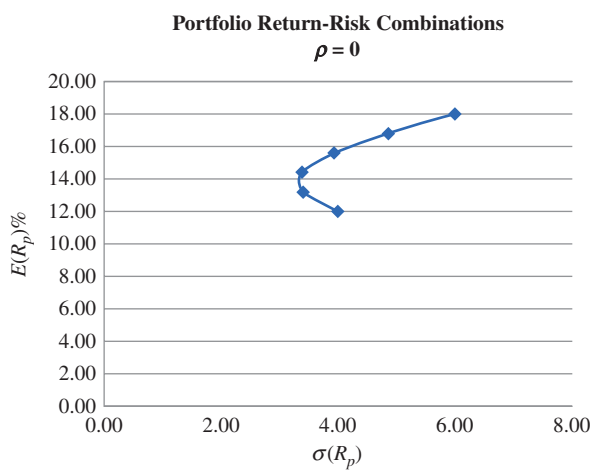


EXHIBIT 8.4 Portfolio Return and Risk with Zero Correlation

Portfolio	w_A	w_B	$E(R_p)$	$\sigma\ (R_p)$
1	1	0	$E(R_{A)} = 12\%$	$\sigma(R_A) = 4$
2	0.8	0.2	13.2%	3.42
3	0.6	0.4	14.4%	3.39
4	0.4	0.6	15.6%	3.93
5	0.2	0.8	16.8%	4.86
6	0	1	$E(R_B) = 18\%$	$\sigma\ (R_B) = 6$



Intuitively, the convex return-risk relation implies that as you move up from the middle of the return-risk graph, you become more specialized in the high return-risk stock. As a result, the portfolio risk takes on progressively more and more of the risk of the high-risk security. In addition, as the portfolio becomes more specialized (and therefore less diversified) it loses the covariance effect. Combined, the increasing proportion allocated to the risky security and the loss of the covariance effect due to specialization causes the portfolio risk to increase at an increasing rate. Since the correlations among many securities are less than one, many portfolio return-risk relations are characterized by this convex relation.

In examining Exhibit 8.3, several additional points should be noted. First, the lower the correlation coefficient, the more dominant the portfolio's return-risk combinations. Thus, for a given risk, over the positively sloped portion of the curves portfolio returns are greater for lower correlation coefficients. Secondly, note that each of the return-risk curves with intermediate correlations has a vertical point (inflection point) where the slope of the curve is zero. This point represents the minimum variance portfolio. For a two-security portfolio, this portfolio can be found with calculus by taking the derivative of the portfolio variance equation with respect to one of the weights, setting the derivative equal to zero, and solving the resulting equation for the weight. Doing this, we obtain:

$$w_{\mathrm{A}} = \frac{\sigma(r_{\mathrm{B}})^2 - \sigma(r_{\mathrm{A}})\sigma(r_{\mathrm{B}})\rho_{\mathrm{AB}}}{\sigma(r_{\mathrm{B}})^2 + \sigma(r_{\mathrm{A}})^2 - 2\sigma(r_{\mathrm{A}})\sigma(r_{\mathrm{B}})\rho_{\mathrm{AB}}}$$

Finally, note that each of the return-risk combinations is unique; that is, each return-risk combination is associated with one allocation. This is because we have limited our analysis to two securities. If we expand our portfolio to more than two securities, then we will find that there are a number of allocations that can yield the same portfolio return and a number of allocations that can yield the same portfolio risk; that is, when the number of securities in a portfolio exceeds two, then the portfolio selection problem is one of determining the allocation that will yield an efficient portfolio—Markowitz portfolio selection.

Markowitz Portfolio Selection

Math Approach

There are several approaches that can be used to solve for the security allocations that satisfy the Markowitz portfolio selection objective. One of these is the mathematical approach and another is quadratic programming. The math approach uses differential calculus to find the allocation that will minimize the portfolio variance subject to the constraints that the weights sum to one and a specified portfolio return is attained, or the allocation that will maximize the portfolio return subject to the constraints that the weights sum to one and a specified portfolio variance is attained.

Thus, for a three-stock portfolio, the objective of portfolio variance minimization would be to solve for the w_1, w_2, and w_3 allocations that would yield the minimum portfolio (Min V_p), subject to the constraints that w_1, w_2, and w_3 sum to one and yield a specified portfolio return, $E_p{}^*$:

Portfolio Inputs:

$$\text{Given Estimates: } V_1, V_2, V_3, C_{12}, C_{13}, C_{23}$$

Objective Function:

$$\text{Minimize } V_p = w_1^2 V_1 + w_2^2 V_2 + w_3^2 V_3 + 2w_1 w_2 C_{12} + 2w_1 w_3 C_{13} + 2w_2 w_3 C_{23}$$

where:

$$V_p = V(R_p)$$
$$V_i = V(r_i)$$
$$C_{ij} = Cov(r_i r_j)$$

First constraint:

$$w_1 + w_2 + w_3 = 1$$

Second constraint:

$$w_1 E_1 + w_2 E_2 + w_3 E_3 = E_p{}^*$$

where:

$$E_i = E(r_i)$$

The math approach for portfolio maximization given a specified portfolio variance is similar to the variance minimization approach. In this case, the objective function is the portfolio expected return, and the constraint is the portfolio variance. The portfolio return maximization approach is consistent with the portfolio variance minimization approach, yielding the same allocations as the portfolio variance approach.

This constrained optimization problem can be solved mathematically using the Lagrangian technique. The approach is presented in Appendix 8A (text Web site) along with an example. The approach requires using matrix algebra to solve a set of equations simultaneously. The Bloomberg Exhibit box, "Using the Bloomberg CORR Screen and Excel to Solve for Markowitz Efficient Portfolios," explains how to download a variance-covariance matrix for a portfolio from its CORR screen to Excel and then explains how to use Excel matrix multiplication commands to solve for a Markowitz

efficient portfolio using the math approach. See the supplemental Appendix B (text Web site) on matrix algebra and for a listing of matrix algebra commands in Excel.

The math approach for solving for Markowitz efficient portfolios is capable of handling large portfolios. Its limitation is that the solutions do not necessarily exclude negative weights. Thus, it is possible to obtain an optimum portfolio that requires taking a short position in a poor security and using the proceeds to invest in other securities in the portfolio. Since most investors do not consider shorting poor securities, the mathematical approach may not be practical.

USING THE BLOOMBERG CORR SCREEN AND EXCEL TO SOLVE FOR MARKOWITZ EFFICIENT PORTFOLIOS

1. The CORR screen can be used to create and save a number of correlation matrices for securities, indexes, currencies, interest rates, and commodities. As noted in Chapter 6, the matrix also shows a variance-covariance matrix (Cov) for portfolios up to 10 stocks.
2. A portfolio created in PRTU can be imported into CORR. To import: (1) Click "Create New" tab; (2) Select dates and period for statistical analysis (e.g., Date Range: 8/5/2006 to 8/8/2013 and weekly periods); (3) in Matrix Securities Box, click "Symmetric Matrix box," and "Add from Sources" tab, Select "Portfolio," Name of Portfolio (e.g., Blue Rock), click "Select All," and click "Update."
3. On the CORR screen, you can obtain the variance-covariance matrix by selecting "Covariance" from the dropdown "Calculations" tab.
4. Matrices in CORR can be exported to Excel by clicking "Export to Excel" in the dropdown "Export" tab in the far right corner of the screen.
5. A *coefficient matrix*, **A**, formed from a variance-covariance matrix and its inverse, $\mathbf{A}^{-1}$, can be used to solve for the Markowitz efficient portfolio; these matrices can be created in Excel.

See Appendix 8A for an explanation of the math approach and Appendix B for a primer on matrix algebra and matrix Excel commands.

See Bloomberg Web Exhibit 8.1.

BLOOMBERG SCREENS FOR THE STOCK RETURNS OF A PORTFOLIO

PORT Screen, Performance Tab

The PORT screen can used to find historical returns of the portfolio and the returns on the stock:

- For portfolio: Performance tab, "Total Return," "Portfolio vs Index" (e.g., Dow (INDU)).
- For stocks: Performance tab, "Main View," "By" dropdown: None, "Portfolio vs None."

- Time = MTD, YTD, and Custom (for Custom, the select time period must be within the period history of the portfolio created in PRTU).
- Note: The Heat Map for the portfolio and stocks can be uploaded by clicking the "Chart" icon on the right.

MRR Screen

The MRR can be used to find average returns for stocks for different time periods. MRR <Enter>; select "Portfolio" from Source tab and your portfolio from the "Name" tab; select the time period for your analysis.

DDM Screen

The DDM estimates a stock's intrinsic value using a three-stage growth model. You can input different values for the lengths of each growth period, growth rates, and discount rates. The DDM also calculated an IRR that can be used as an estimate of expected rate. See Bloomberg exhibit box in Chapter 3.

See Bloomberg Web Exhibit 8.2.

Quadratic Programming Approach

An alternative to the mathematical approach is quadratic programming (QP). QP is an algorithm that iteratively solves for the security weights that yield the minimum portfolio variance subject to three constraints: The weights sum to one, they yield a specified portfolio expected return, and each weight is nonnegative. For a three-stock portfolio, the QP approach can be defined as follows:

Portfolio Inputs:

$$\text{Given Estimates: } V_1, V_2, V_3, C_{12}, C_{13}, C_{23}$$

Objective Function:

$$\text{Minimize: } V_p = w_1^2 V_1 + w_2^2 V_2 + w_3^2 V_3 + 2w_1 w_2 C_{12} + 2w_1 w_3 C_{13} + 2w_2 w_3 C_{23}$$

First constraint:

$$w_1 + w_2 + w_3 = 1$$

Second constraint:

$$w_1 E_1 + w_2 E_2 + w_3 E_3 = E_p^*$$

Third constraint of nonnegative weights:

$$w_1 \geq 0; \quad w_2 \geq 0; \quad w_3 \geq 0$$

With the nonnegative weight constraints, quadratic programming provides a more practical approach to constructing portfolios that satisfy the Markowitz objective.

Excel Solver Approach: Bloomberg's Asset Allocation Optimizer Template

In Excel, efficient portfolios (maximum E_p* given V_p or minimum V_p* given E_p) with nonnegative weight constraints can be generated using the Excel Solver Add-In. Such programs yield results similar to QP-generated portfolios. A Bloomberg Excel program that uses Excel Solver for solving for such portfolios, "Asset Allocation Optimizer," can be downloaded from the Bloomberg Excel Template library found on the DAPI screen (DAPI <Enter>; Click Excel Template Library and Portfolios topic and then select "Asset Allocation Optimizer.") The template is shown in Exhibit 8.5. To use the program, the user:

1. Inputs the stock tickers (for stocks, ticker with the "Equity" moniker, for indexes, ticker with "Index" moniker, etc.).
2. Inputs average returns or expected returns.
3. Selects the time period for calculating the variance-covariance matrix (the template in Exhibit 8.5 shows the correlation matrix).
4. Selects minimum weight and maximum weight constraints for each stock; here the user can set the minimum weights to zero and the maximum weights at 99 percent (or another specified constraint).
5. Inputs the risk-free rate and values for the optimization programs: the portfolio standard deviation for portfolio maximization optimization and portfolio return for portfolio variance minimization.
6. Clicks "Optimize Weights" tab to run the program.

Exhibit 8.5 shows the optimum solutions for the 10 stocks making up the Blue Rock Fund. The variance-covariance matrix is calculated for the time period from 8/5/2006 to 8/8/2013 (weekly prices are used), the expected stock returns are based on averages over a more recent time period, the specified portfolio return is 15.5 percent, and the specified annualized portfolio standard deviation is 17 percent.[2] As shown at the bottom of the table in Exhibit 8.5, the optimum portfolio weights for the variance minimization are 30.7 percent in CVS, 21.4 percent in Disney, 22 percent in Johnson & Johnson, and 25.9 percent in Kroger.

The ex-post total returns from the one-year period from 8/13/12 to 8/13/13 and for the three-year period from 8/13/10 to 8/13/13 for the Markowitz efficient portfolio and the S&P 500 are shown in Exhibit 8.6 (Bloomberg's PORT screen, Performance tab, and Total Return tab). As shown in the exhibit, the back testing results show the portfolio out-performs the market for both the one-year period (total return of 44.82 percent compared to 16.11 percent total return for S&P 500) and the three-year period (total return of 96.22 percent compared to 56.03 percent total return for S&P 500). Using the Bloomberg PORT screen (Performance tab and Statistics Summary tab) the

EXHIBIT 8.5 Markowitz Efficient Portfolios Generated Using the Asset Allocation Optimizer Excel Program

Asset Allocation Optimizer

Bloomberg

This application requires Excel's Solver Add-in to be installed. Go to Help tab for directions. You may also contact Bloomberg Help Desk.

Asset Allocation Optimizer uses either historical returns or user-customized forecasted returns to generate optimal portfolios.

Follow directions on the left side of the screen to start using the application. You may customize start and end dates for historical return, standard deviation, and corr

1) Enter Tickers ---> 2) Enter Asset Class ---> 3) Choose Return Type

Type 1: Historical

Type 2: Forecasted

Tickers:	Adm Equity	AFL Equity	CVS Equity	Dis Equity	Duk Equity	JNJ Equity	KR Equity	MSFT Equity	PG Equity	XOM Equity
Asset Class:	ADM	AFL	CVS	Dis	Duke	J&J	Kroger	Microsoft	PG	Exxon

	ARCHER-DANIELS-	AFLAC INC	CVS CAREMARK	WALT DISNEY CO/THE	DUKE ENERGY CORP	JOHNSON & JOHNSON	KROGER CO	MICROSOFT CORP	PROCTER & GAMBLE	EXXON MOBIL CORP
Returns *	0.61%	13.61%	16.91%	19.78%	7.57%	9.07%	15.77%	7.01%	6.99%	5.18%
Standard Dev	32.5%	43.3%	24.7%	27.6%	18.5%	15.9%	24.2%	27.1%	17.3%	22.1%

* For demonstration only; these are not recommendations; please review your inputs carefully.

You have chosen forecasted rates. Please go to the Forecasted Rates Tab to review your return assumptions.

4) Enter Dates Below

Start Date: 8/5/2006

End Date: 8/8/2013

Historical returns, correlations and standard deviations will update according to dates chosen.

	ARCHER-DANIELS-	AFLAC INC	CVS CAREMARK	WALT DISNEY CO/THE	DUKE ENERGY CORP	JOHNSON & JOHNSON	KROGER CO	MICROSOFT CORP	PROCTER & GAMBLE	EXXON MOBIL CORP
ARCHER-	1.000									
AFLAC INC	0.366	1.000								
CVS CAREMARK	0.231	0.300	1.000							
WALT DISNEY	0.471	0.615	0.389	1.000						
DUKE ENERGY	0.347	0.438	0.357	0.481	1.000					
JOHNSON &	0.431	0.458	0.254	0.560	0.589	1.000				
KROGER CO	0.314	0.279	0.290	0.350	0.369	0.444	1.000			
MICROSOFT	0.233	0.400	0.311	0.477	0.401	0.433	0.298	1.000		
PROCTER &	0.350	0.414	0.263	0.555	0.496	0.585	0.381	0.365	1.000	
EXXON MOBIL	0.564	0.444	0.374	0.604	0.506	0.542	0.395	0.478	0.498	1.000

5) Review Constraints ---> ☑ Constraints Kept

Min Weight	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Max Weight	99.0%	99.0%	99.0%	99.0%	99.0%	99.0%	99.0%	99.0%	99.0%	99.0%

6) Press Button - Optimize

Optimize Weights

Objective 1: Portfolio that minimizes risk				Risk Free Return:	1.50%		Return:	9.80%	Standard Dev:	13.8%
Weights	0.0%	0.0%	14.6%	0.0%	13.9%	38.6%	5.4%	0.2%	27.3%	0.0%
Objective 2: Portfolio that maximizes return				Risk Free Return:	1.50%		Return:	19.75%	Standard Dev:	27.4%
Weights	0.0%	0.0%	1.0%	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Objective 3: Portfolio that maximizes Sharpe Ratio				Risk Free Return:	1.50%		Return:	17.47%	Standard Dev:	19.1%
Weights	0.0%	0.0%	33.9%	33.1%	0.0%	0.3%	32.7%	0.0%	0.0%	0.0%
Objective 4: Portfolio that maximizes return (*given a volatility)				Risk Free Return:	2.50%		Return:	15.53%	Standard Dev:	17.0%
Weights	0.0%	0.0%	31.1%	21.4%	0.0%	21.6%	26.0%	0.0%	0.0%	0.0%
Objective 5: Portfolio that minimizes risk (*given a return)				Risk Free Return:	2.50%		Return:	15.50%	Standard Dev:	17.0%
Weights	0.0%	0.0%	30.7%	21.4%	0.0%	22.0%	25.9%	0.0%	0.0%	0.0%

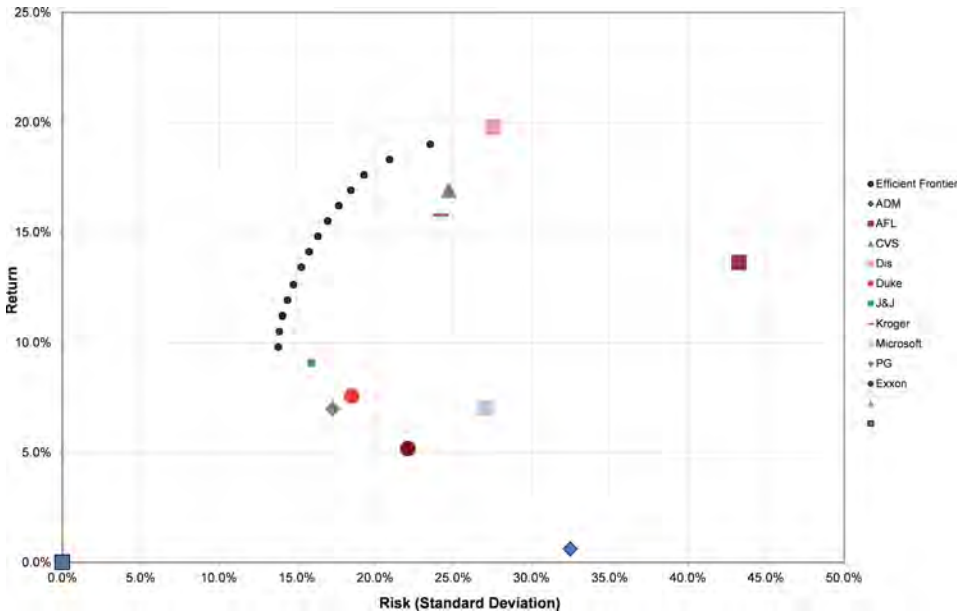
(a)

Risk	Return	ADM	AFL	CVS	Dis	Duke	J&J	Kroger	Microsoft	PG	Exxon
13.8%	9.8%	0.0%	0.0%	14.6%	0.0%	13.9%	38.6%	5.4%	0.2%	27.3%	0.0%
13.9%	10.5%	0.0%	0.0%	18.3%	0.0%	10.2%	39.5%	9.2%	0.0%	22.8%	0.0%
14.1%	11.2%	0.0%	0.0%	22.2%	0.0%	6.4%	39.5%	13.1%	0.0%	18.8%	0.0%
14.4%	11.9%	0.0%	0.0%	24.4%	2.5%	4.2%	38.4%	15.3%	0.0%	15.1%	0.0%
14.8%	12.6%	0.0%	0.0%	25.6%	5.8%	2.1%	37.6%	17.5%	0.0%	11.4%	0.0%
15.3%	13.4%	0.0%	0.0%	27.4%	9.7%	0.0%	35.8%	19.5%	0.0%	7.7%	0.0%
15.8%	14.1%	0.0%	0.0%	28.3%	13.2%	0.0%	33.8%	21.8%	0.0%	2.9%	0.0%
16.3%	14.8%	0.0%	0.0%	29.7%	17.2%	0.0%	29.8%	23.2%	0.0%	0.0%	0.0%
17.0%	15.5%	0.0%	0.0%	30.7%	21.4%	0.0%	22.0%	25.9%	0.0%	0.0%	0.0%
17.7%	16.2%	0.0%	0.0%	31.9%	25.6%	0.0%	14.3%	28.2%	0.0%	0.0%	0.0%
18.5%	16.9%	0.0%	0.0%	33.1%	29.8%	0.0%	6.6%	30.4%	0.0%	0.0%	0.0%
19.3%	17.6%	0.0%	0.0%	33.4%	36.1%	0.0%	0.0%	30.4%	0.0%	0.0%	0.0%
20.9%	18.3%	0.0%	0.0%	28.3%	55.3%	0.0%	0.0%	16.4%	0.0%	0.0%	0.0%
23.5%	19.0%	0.0%	0.0%	23.2%	74.2%	0.0%	0.0%	2.6%	0.0%	0.0%	0.0%
27.4%	19.8%	0.0%	0.0%	1.0%	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

(b)

(*Continued*)

EXHIBIT 8.5 (*Continued*)



(c)

Markowitz portfolio has a beta close to one based on year-to-date calculation, as well as a very large alpha, suggesting abnormal returns.

Elton, Gruber, and Padberg also have developed an algorithm based on the single-index model (examined later) for determining the best efficient portfolio. The technique they derive for generating efficient portfolios, in turn, can also be constrained so that the weights are nonnegative. The technique is presented in Appendix 8B (text Web site) and its application is presented later in this chapter.

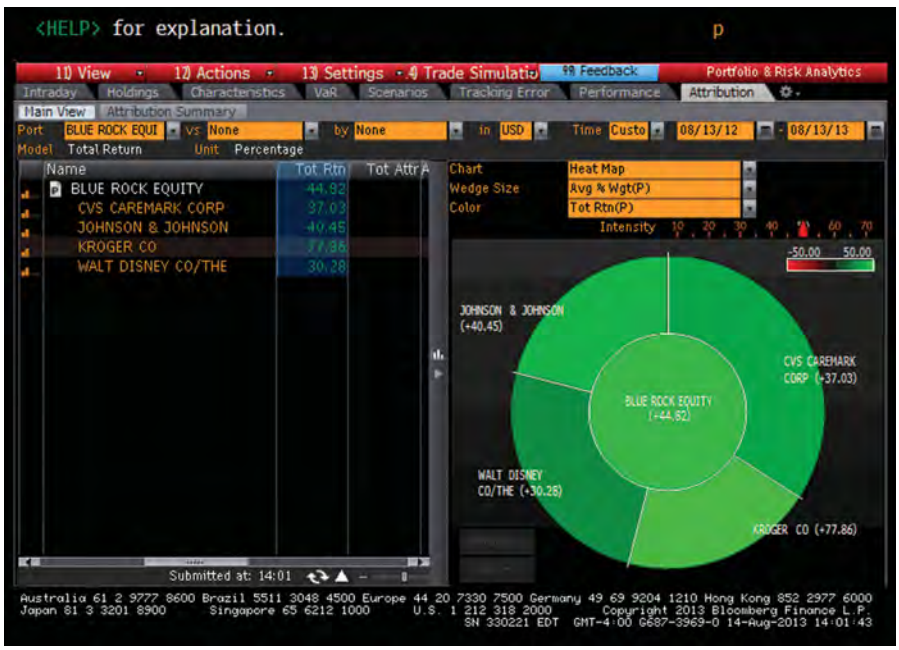
Efficiency Frontier

The Markowitz portfolio selection objective can be restated as one of deriving an efficiency frontier, EF. An efficiency frontier is a graph showing the portfolio expected return, $E(R_p)$, and standard deviation, $\sigma(R_p)$, combinations that are Markowitz efficient; that is, satisfy the Markowitz objective of maximum $E(R_p)$ given a specified $V(R_p)$, or minimum $V(R_p)$ given a specified $E(R_p)$.

There are three steps involved in generating an efficiency frontier. The first step is to estimate the portfolio inputs: $E(r_i)$, $V(r_i)$, and $Cov(r_i\ r_j)$. These parameters can be estimated using either historical averages or a regression model. There are several regression models that can be used. The simplest is the single-index model in which each security's return is regressed against the market return. As discussed in the next section, the single-index model reduces the number of estimates needed to generate

EXHIBIT 8.6 Ex-Post Performance, 8/13/2012–8/12/13 Markowitz Portfolio (Blue Rock Equity) and S&P 500

(a)

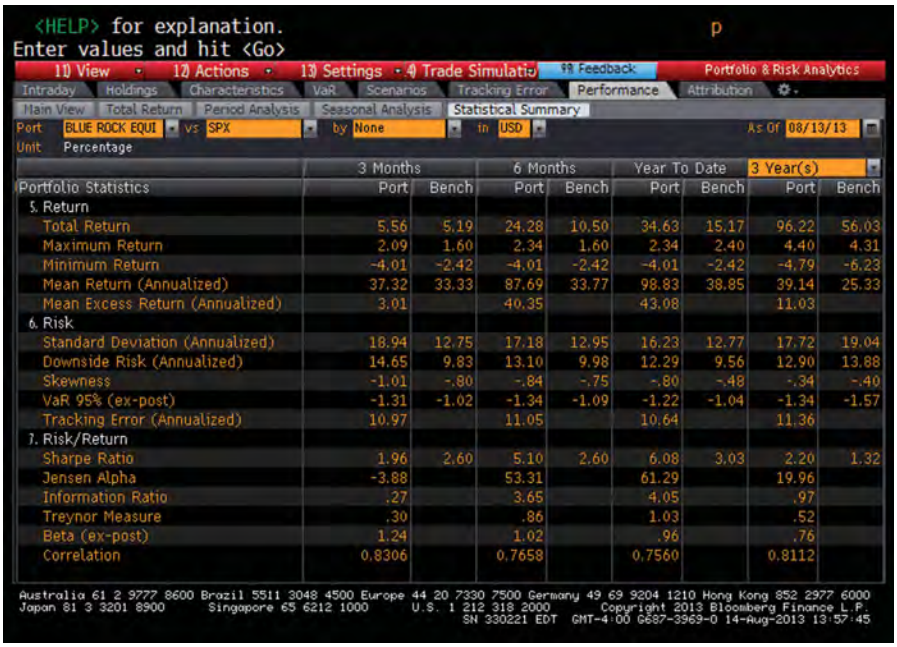


(b)

(*Continued*)

EXHIBIT 8.6 (*Continued*)

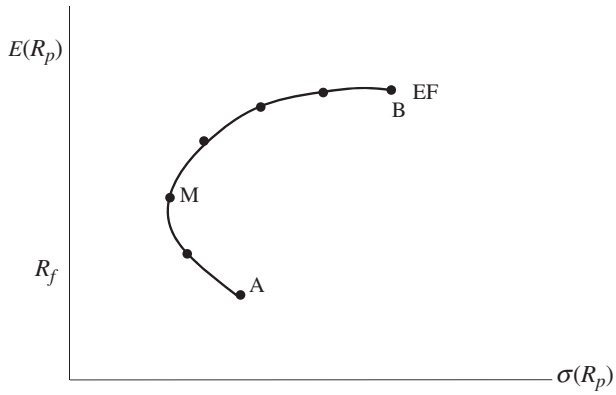
(c)



	3 Months		6 Months		Year To Date		3 Year(s)	
Portfolio Statistics	Port	Bench	Port	Bench	Port	Bench	Port	Bench
5. Return								
Total Return	5.56	5.19	24.28	10.50	34.63	15.17	96.22	56.03
Maximum Return	2.09	1.60	2.34	1.60	2.34	2.40	4.40	4.31
Minimum Return	-4.01	-2.42	-4.01	-2.42	-4.01	-2.42	-4.79	-6.23
Mean Return (Annualized)	37.32	33.33	87.69	33.77	98.83	38.85	39.14	25.33
Mean Excess Return (Annualized)	3.01		40.35		43.08		11.03	
6. Risk								
Standard Deviation (Annualized)	18.94	12.75	17.18	12.95	16.23	12.77	17.72	19.04
Downside Risk (Annualized)	14.65	9.83	13.10	9.98	12.29	9.56	12.90	13.88
Skewness	-1.01	-.80	-.84	-.75	-.80	-.48	-.34	-.40
VaR 95% (ex-post)	-1.31	-1.02	-1.34	-1.09	-1.22	-1.04	-1.34	-1.57
Tracking Error (Annualized)	10.97		11.05		10.64		11.36	
7. Risk/Return								
Sharpe Ratio	1.96	2.60	5.10	2.60	6.08	3.03	2.20	1.32
Jensen Alpha	-3.88		53.31		61.29		19.96	
Information Ratio	.27		3.65		4.05		.97	
Treynor Measure	.30		.86		1.03		.52	
Beta (ex-post)	1.24		1.02		.96		.76	
Correlation	0.8306		0.7658		0.7560		0.8112	

(d)

EXHIBIT 8.7 Efficiency Frontier

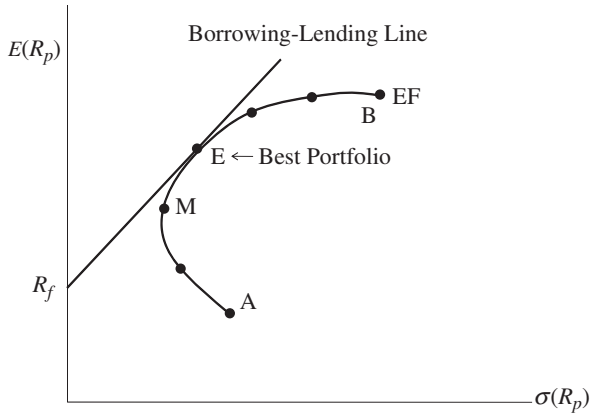


efficient portfolios, and it has been shown to lead to better estimates of the portfolio inputs than historical averages. Some practitioners also use a multi-index model in which each security's return is regressed against several explanatory variables.

Once the expected returns, variances, and covariances of the securities included in the portfolio are estimated, the next step is to generate Markowitz efficient portfolios. As noted above, this can be done using either the math approach, quadratic programming, or with Excel Solver. With any of these approaches, one would first specify a number of portfolio expected returns (or variances), then solve for the weights that would yield the minimum portfolio variance (or maximum portfolio return) for each return (or variance). Each portfolio return and variance would be either a Markowitz efficient or inefficient portfolio. The last step is to plot each portfolio's expected return and standard deviation (not portfolio variance) to generate the efficiency frontier.

The EF curve shown in Exhibit 8.7 depicts a typical efficiency frontier. The curve has features similar to the return-risk graph for a two-security portfolio previously discussed. First, like the two-security return-risk graph, EF is characterized by both a negatively sloped portion and positively sloped portion. The negatively sloped portion of EF represents the inefficient portfolios [minimum $E(R_p)$, given $\sigma(R_p)$], whereas the positively-sloped portion shows the efficient portfolios. Second, the efficiency frontier, like the portfolio return and risk curve for a two-security portfolio, is convex from below, except for cases in which the securities are perfectly positively or negatively correlated. As discussed with the two-security portfolio, the convexity of the efficiency frontier is explained by the increase specialization in the high-risk security and the loss of the covariance effect that occurs as one moves up the efficiency frontier. Finally, the efficiency frontier has a vertical segment (inflection point) that defines the minimum variance portfolio. The efficiency frontier and table for the portfolio in Exhibit 8.5 generated by using the Bloomberg "Asset Allocation Optimizer" is shown in Exhibit 8.5c.

EXHIBIT 8.8 Best Efficiency Portfolio



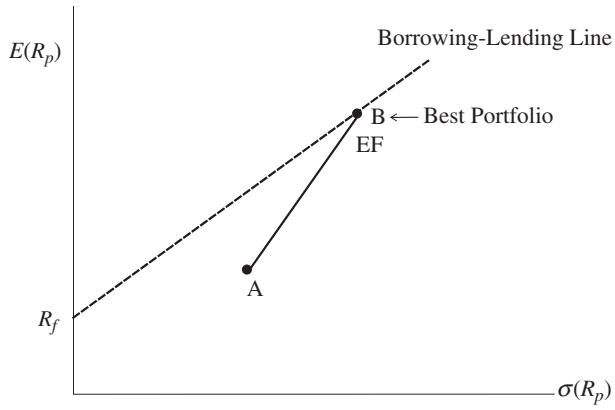
Best Efficient Portfolio

All portfolios along the positively sloped portion of the efficiency frontier are Markowitz efficient; that is, all have the maximum $E(R_p)$ given $V(R_p)$, or minimum $V(R_p)$ given $E(R_p)$. From this set of efficient portfolios, it is possible to determine the best portfolio by using the concept of portfolio ranking examined in Chapter 7. This can be seen in Exhibit 8.8, in which portfolio E represents the best portfolio. This is because the borrowing and lending line constructed with portfolio E has the steepest slope (largest λ). Thus, the return-risk combinations available with portfolio E and a risk-free security dominate the return-risk opportunities available from any other efficient portfolio and the risk-free security. If the efficiency frontier is convex (as we expect for many portfolios), then the best portfolio, such as E, can be determined at the point of tangency of the efficiency frontier and the borrowing and lending line. Furthermore, if the efficiency frontier is convex, then the best portfolio would be defined as one of the middle points on EF (not at a corner), implying that the best portfolio would be diversified.

Efficiency Frontiers for Stocks with Perfect Positive and Negative Correlations

It should be noted that if the securities in the portfolio are perfectly positively correlated, then the efficiency frontier is linear and the best portfolio will be one that is at one of the corners. Since the corner points of an efficiency frontier define a one-security portfolio (either the low return-risk security or the high return-risk security), the best portfolio therefore consists of only one security (see Exhibit 8.9). Thus, if the securities in the portfolio are perfectly positively correlated, then there is no benefit to diversification and the best portfolio consists of either the low return-risk security or the high return-risk security. This observation confirms a previously stated observation that if securities are moving in unison, there is no diversification benefit and

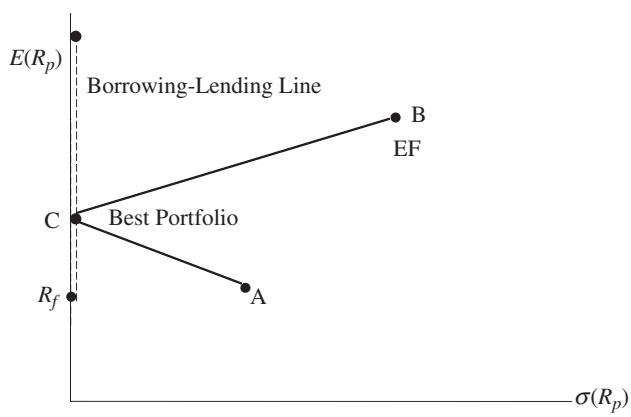
EXHIBIT 8.9 Efficiency Frontier with Perfect Positive Correlation



a portfolio with a large number of perfectly positively correlated securities would be superfluous.

Finally, note that if the securities are perfectly negatively correlated, then the best portfolio (i.e., the one with the steepest borrowing-lending line) would be the zero risk portfolio obtained with limited diversification in two securities that are perfectly negatively correlated (see Exhibit 8.10). In this case, if the securities are in fact perfectly negatively correlated, or if two perfectly negatively correlated positions are formed (e.g., long stock position and a short stock position formed with stock option positions), then an arbitrage opportunity would exist if the risk-free rate were different than the portfolio rate associated with the zero risk portfolio. For example, if the risk-free rate were less than the zero-risk portfolio's rate, then an arbitrageur would borrow as much as she could to invest in the portfolio. By doing this, the arbitrageur would realize a free lunch: a future dollar return with no risk and no investment—an arbitrage.

EXHIBIT 8.10 Efficiency Frontier with Perfect Negative Correlation



BLOOMBERG'S ASSET ALLOCATION OPTIMIZER

To access Asset Allocation Optimizer Excel template go to the template library found on the DAPI screen (DAPI <Enter>) and click "Excel Template Library," "Equity," and "Portfolios." The template is shown in Exhibit 8.5.

As described in the Template's "Help" sheet, the optimization program uses historical returns or user-customized forecasted returns to generate optimal portfolios. You can customize beginning and ending dates for the historical returns, standard deviation, and correlation matrix data. The spreadsheet uses Microsoft Excel's Solver Add-in to solve portfolio equations to find optimal portfolios. Instructions are provided in Help for uploading the Add-in if it is not already on your Excel spreadsheet. The program generates an optimal portfolio in the "Optimizer tab" and builds an efficient frontier in the "Efficient Frontier" tab. For an example of how to use the template, see Exhibit 8.5, "Markowitz Efficient Portfolios Generated Using the Asset Allocation Optimizer Excel Program."

See Bloomberg Web Exhibit 8.3.

Single-Index Model

Markowitz Portfolio Theory was introduced over 60 years ago. Since then most of the academic research on Markowitz portfolio analysis has focused on how to implement the theory. One area that initially drew the interest of researchers was the simplification of the computational process. Recall that a portfolio with 100 stocks would require estimates of 4,950 covariances, 100 variances, and 100 expected returns. Two models that reduce the number of calculations needed to generate efficient portfolios are the *single-index model* and the *multi-index model*.

Developed by William Sharpe, the single-index model (also called the *diagonal model*) assumes that all securities in a portfolio are related just to the market return and that there is no correlation between the unsystematic risks of securities. Combined, these assumptions imply that comovements between securities in a portfolio are related to a single factor—the market return. As a result, in the single-index model one does not have to estimate the correlations between stocks in the portfolio; instead, one only has to estimate each security's relation to the common factor. In Chapter 6, we examined this model when we looked at estimating stock's expected return, variance, and covariance. Formally, the single-index model assumes that each security i in the portfolio being evaluated is only related to the market as described by the regression model:

$$r_i = \alpha_i + \beta_i R^M + \varepsilon_i \tag{8.1}$$

The model also assumes that the standard regression assumptions hold for each security in the portfolio and that there is no correlation between the error terms of